Organic and Sustainable Farming Practices

Agriculture RAG Knowledge Base • Educational reference document

Sustainable farming seeks to maintain productive agriculture while protecting soil, water, biodiversity, and the long-term viability of farming systems. Organic agriculture is a specific production system governed by standards and restrictions on permitted inputs; sustainable agriculture is a broader concept and does not necessarily mean organic certification.

Practices associated with sustainable soil management include crop rotations, cover crops, residue management, erosion control, compost or other suitable organic amendments, and efficient nutrient use. These practices can contribute to soil structure, nutrient cycling, and water management.

Organic systems rely heavily on biological and cultural processes for fertility and pest management. Nutrient sources can include compost, manure, crop residues, and approved mineral inputs depending on the applicable organic standard. Nutrient availability must still be considered carefully because organic sources vary in nutrient concentration and release rate.

Sustainable agriculture also includes efficient use of water, responsible pesticide management, biodiversity conservation, and reduction of unnecessary input losses. Economic sustainability matters too: practices must be feasible for farmers in terms of labor, equipment, cost, risk, and market access.

Good agricultural decisions are context-dependent. A recommendation should consider the crop, soil, climate, farm resources, and applicable local regulations or standards.

Reference basis

• FAO, Organic Agriculture resources

• FAO, Integrated Plant Nutrient Management

Note: This document is a project knowledge-base document prepared for a student RAG demonstration. It is not a substitute for crop-specific agronomic advice, soil testing, product labels, or local agricultural recommendations.